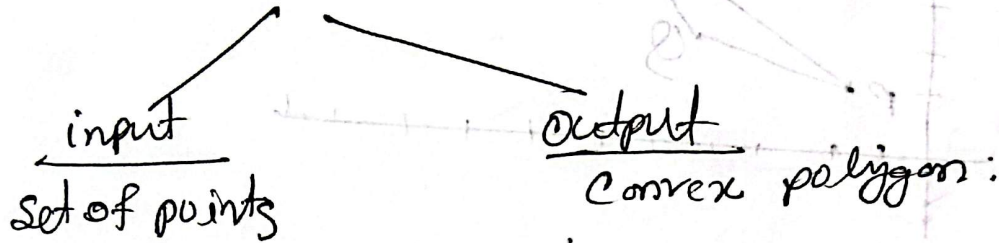


Computational Geometry

Convex Hull



interior angle $\geq 180^\circ$ or $\geq \pi$ or $\geq 2\pi$

$$\vec{PQ} \times \vec{PR} = \begin{vmatrix} 3 & 1 \\ 4 & 4 \end{vmatrix} = 3 \times 4 - 1 \times 4 = 8 > 0$$

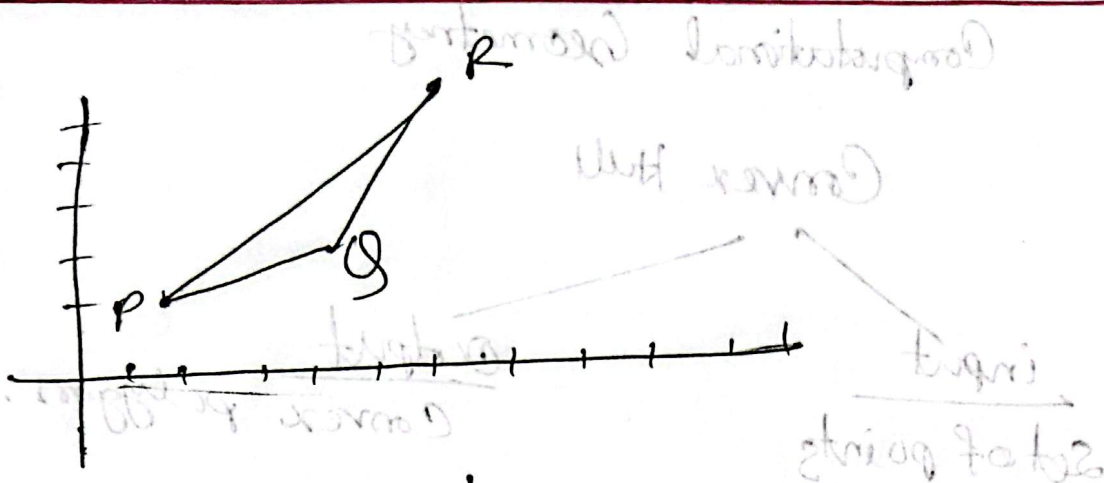
Angle $\geq 180^\circ$ or $\geq \pi$ or $\geq 2\pi$, point Q is not in the convex hull. Counter-clockwise order.

$P(1,1)$ $Q(4,2)$ $R(5,5)$

$\vec{PQ} = (3,1)$
 $\vec{PR} = (4,4)$

$\vec{PQ} \times \vec{PR} = \begin{vmatrix} 3 & 1 \\ 4 & 4 \end{vmatrix} = 8 > 0$ left turn.

Counter-clockwise



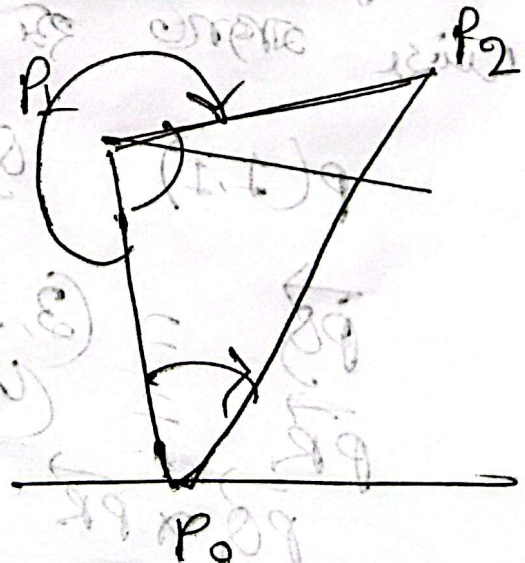
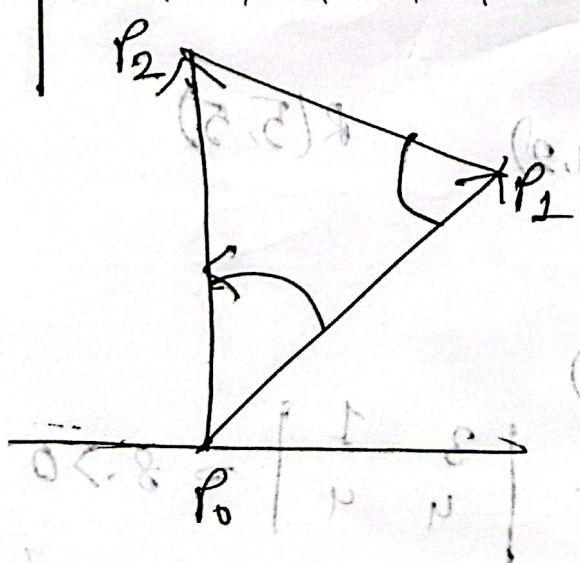
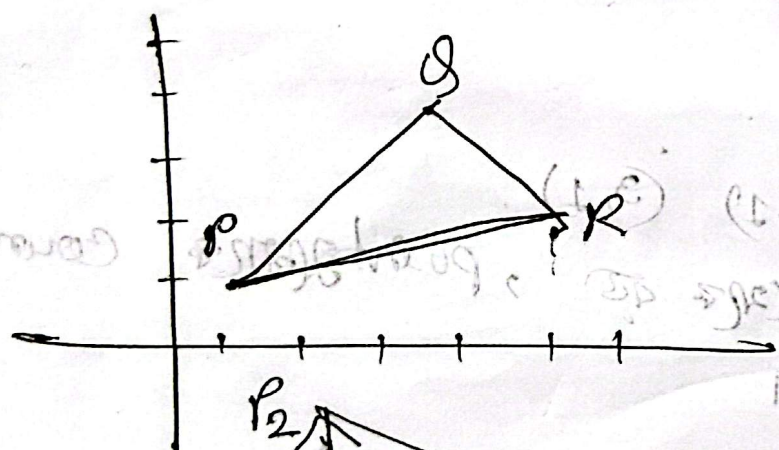
$P(1,1), Q(4,4), R(5,2)$.

$$\vec{PQ} = (3, 3)$$

$$\vec{PR} = (4, 1)$$

$$\vec{PQ} \times \vec{PR} = \begin{bmatrix} 3 & 3 \\ 4 & 1 \end{bmatrix} = 3 - 12 = -9 < 0$$

right Turn.



left turn $\angle < 180^\circ$
 right turn $\angle > 180^\circ$

✓ $\rightarrow \times \rightarrow \times \rightarrow$
 $A(0,0), B(2,1), C(1,3), D(4,4), E(5,1), F(2,2)$

A $\rightarrow$ sort $\rightarrow$ polar angle



complexity = $n \log n$

Graham Scan

counterclockwise

sort $A(0,0), C(1,3), B(2,1), F(2,2), D(4,4), E(5,1)$
 $0^\circ, 71.56^\circ, 26.56^\circ, 45^\circ, 45^\circ, 11.31^\circ$

F D no problem.

A E B D F C

$$\begin{vmatrix} -1 & -3 \\ 3 & 1 \end{vmatrix} = -1 + 9 = 8$$

E, B, D.

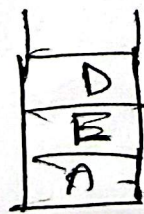
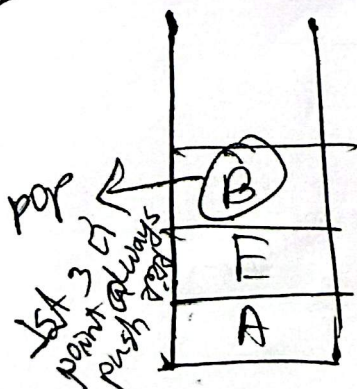
$$\vec{EB} \times \vec{ED} = (3,0) \times (1,-3)$$

$$= \begin{vmatrix} 3 & 0 \\ 1 & -3 \end{vmatrix} = -9$$

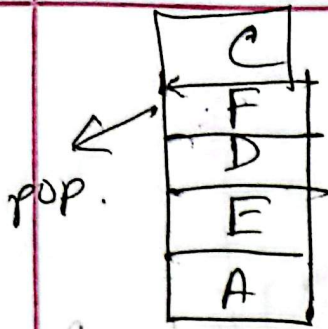
= right turn

$$E, D, F \quad \vec{ED} \times \vec{EF} = \begin{vmatrix} -1 & -3 \\ 3 & 1 \end{vmatrix} = -1 + 9 = 8$$

left turn



3
0
= -2



EDF
EDXEF

D, F, C.

DF x DC = ...

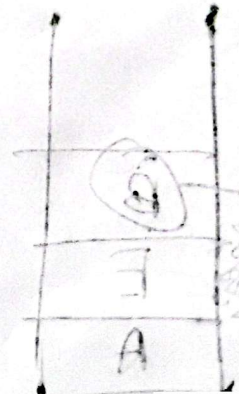


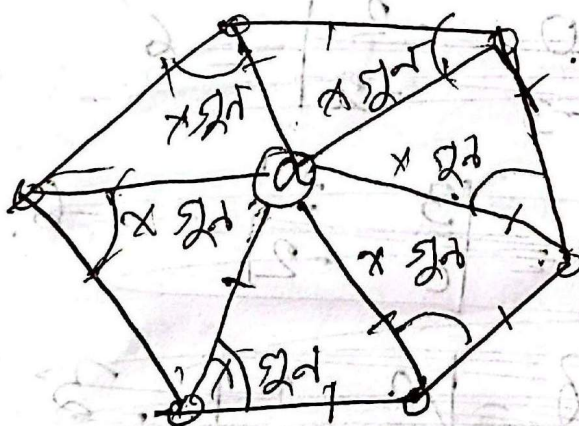
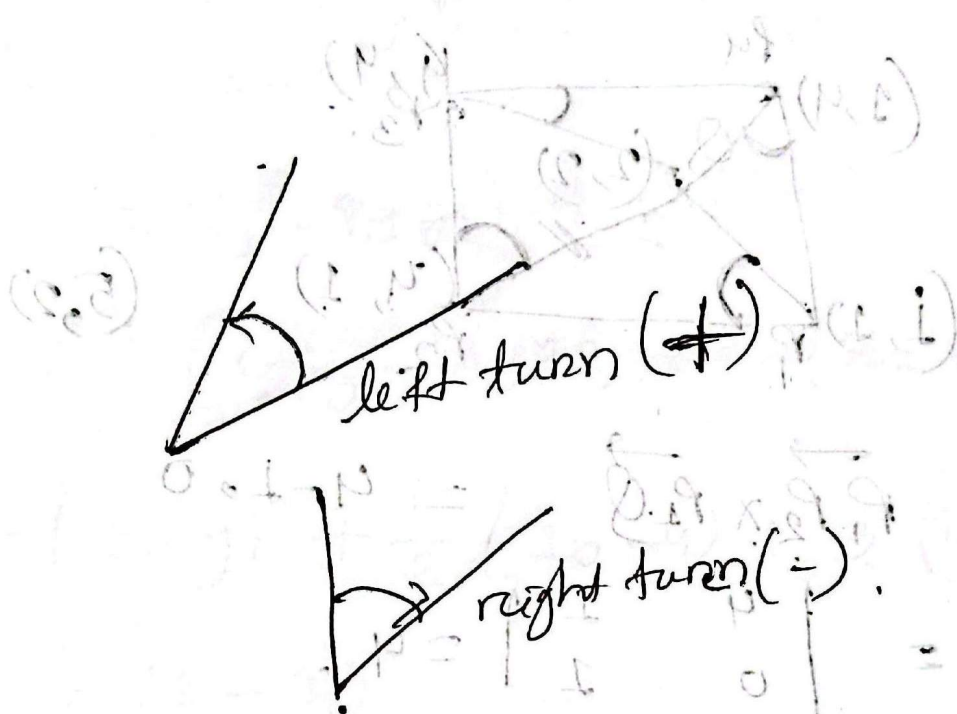
Jarvis algorithm

nearest 2 point
line segment

Convex Hull.

↓



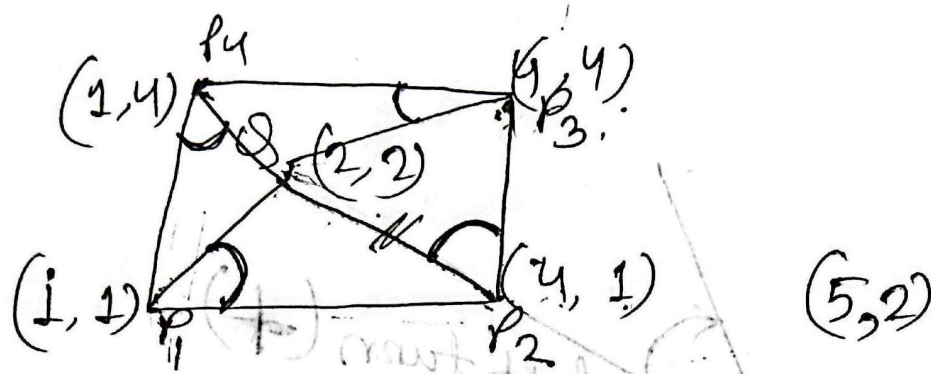


Shoaley's formula.

point is inside or outside
area is 1

অবস্থানটি কেন্দ্র? যদিও অক্ষর 0 1

outside



$$\vec{P_1P_2} \times \vec{P_2P_3} = 4 - 1 = 0$$

$$= \begin{vmatrix} 4 & 1 \\ 0 & 1 \end{vmatrix} = 4$$

~~$$\vec{P_3P_2} \times \vec{P_2P_1} = \begin{vmatrix} 0 & -2 \\ -3 & -2 \end{vmatrix} = 6$$~~

~~$$\vec{P_4P_3} \times \vec{P_3P_2} = \begin{vmatrix} 3 & 1 \\ 0 & -2 \end{vmatrix} = -6$$~~

~~$$\vec{P_1P_4} \times \vec{P_4P_3} = \begin{vmatrix} 0 & 1 \\ -3 & -2 \end{vmatrix} = 3$$~~

$$\vec{P_2P_3} \times \vec{P_3P_4}$$

$$\vec{P_3P_4} \times \vec{P_4P_1}$$

$$\vec{P_4P_1} \times \vec{P_1P_2}$$

এখানে 0 হলো
 inside থাকিলে
 check না করলে
 হবে।

$$\begin{matrix} (1, 1) & (4, 2) & (4, 4) & (1, 4) & (1, 1) \\ (x_0, y_0) & (x_1, y_1) & (x_2, y_2) & (x_3, y_3) & (x_4, y_4) \end{matrix}$$

$$\begin{aligned} \text{Area} &= \frac{1}{2} \sum_{i=0}^{n-1} (x_i y_{i+1} - x_{i+1} y_i) \\ &= \frac{1}{2} (x_0 y_1 - x_1 y_0) + \frac{1}{2} (x_1 y_2 - x_2 y_1) + \\ &\quad + \frac{1}{2} (x_2 y_3 - x_3 y_2) + \frac{1}{2} (x_3 y_4 - x_4 y_3) \\ &= \frac{1}{2} (1 \cdot 4) + \frac{1}{2} (16 - 4) + \frac{1}{2} (16 - 4) + \\ &\quad \frac{1}{2} (1 - 4) \\ &= \frac{1}{2} (-3) + \frac{1}{2} (12) + 6 + \frac{1}{2} (-3) \\ &= 12 - \frac{3}{2} + \frac{3}{2} = 12 - 0 = 12 \\ &= 12 - \left(\frac{3}{2} + \frac{3}{2} \right) = 12 - 3 = 9 \end{aligned}$$

Number Theory

equivalent

$$\left. \begin{aligned} &\text{gcd}(a, b) \\ &\equiv \text{gcd}(b, a \bmod b) \end{aligned} \right\}$$

$$\begin{array}{r} 2 \overline{) 110} \\ \underline{515} \\ 1 \end{array} \quad \begin{array}{r} 2 \overline{) 12} \\ \underline{26} \\ 313 \end{array}$$

$$\begin{aligned} &\text{gcd}(10, 12) \\ &= \text{gcd}(12, 10) \\ &= 2 \end{aligned}$$